Royal University of Bhutan

College of Science and Technology

LAB REPORT (Practical 6)

STANDARD ACCESS LIST CONFIGURATION

Name: Kencho Dorji

Student ID: 02250353

Department: Software Engineering

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LAB 6. Standard Access List Configuration

A. Aim:

To establish standard access list in a LAN and configure between the networks.

B. Objective:

Configure between the networks in a Local Area Network (LAN) and insert standard access list in a router. Whereby it decides to allow a client to send data to other networks, or deny it's access by configuring in a router. It also let us understand on how standard access list is applied and it's working within or between the networks.

C. Background:

An Access Control List (ACL) is a list of rules that control and filter traffic based on source and destination IP addresses or Port numbers. This happens by either allowing packets or blocking packets from an interface on a router, switch, firewall etc.

In addition to traffic control and filtering in a network, ACL can also be used as a security measure for connecting to your router by allowing only the necessary IP addresses or networks for accessing the router.

- **<u>Standard Access Lists</u>**

Standard access lists are the basic form of access list on Cisco routers that can be used to match packets by source IP address field in the packet header. These access lists are simpler to create and understand but packet matching options are also limited to only source address. IP Standard Access Lists 1-99

Some functions are:

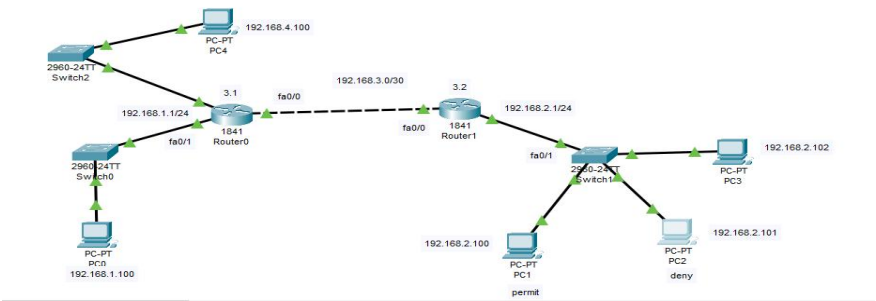
- ✓ Traffic Filtering.
- ✓ Network Security
- ✓ Bandwidth management
- ✓ Traffic Flow Management

D. Procedure:

1. Devices Required:

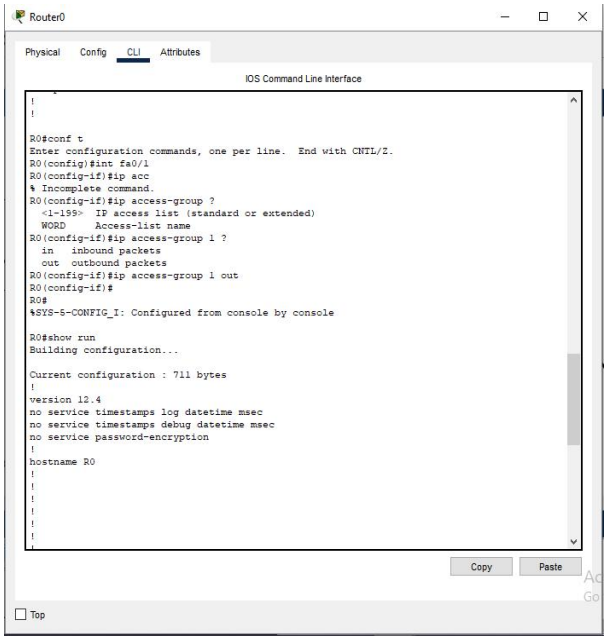
- ✓ Two Routers
- ✓ Three switches
- ✓ Five PCs

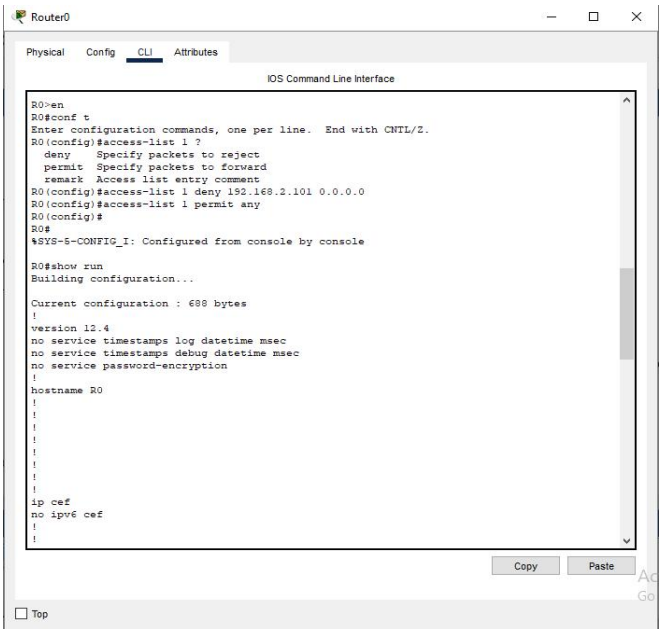
Connect with appropriate cables and configure them correctly.



2. Assign Standard Access List:

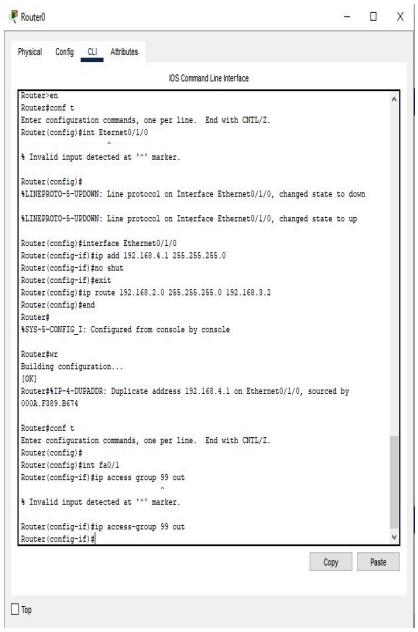
- ✓ Go to router 0 and assign the standard list in Interface FastEthernet0/1.
- ✓ Set it as outbound from fa0/0

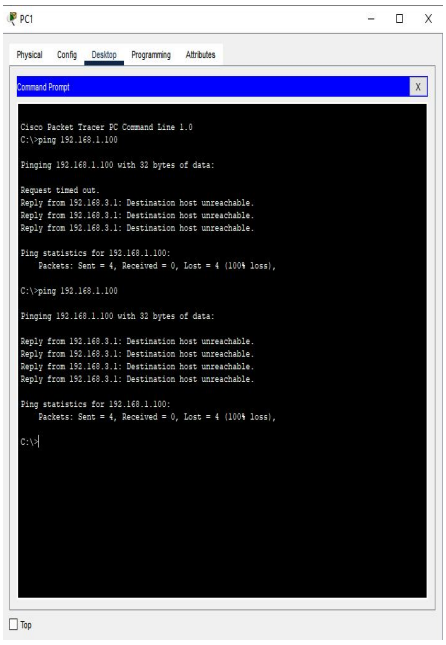


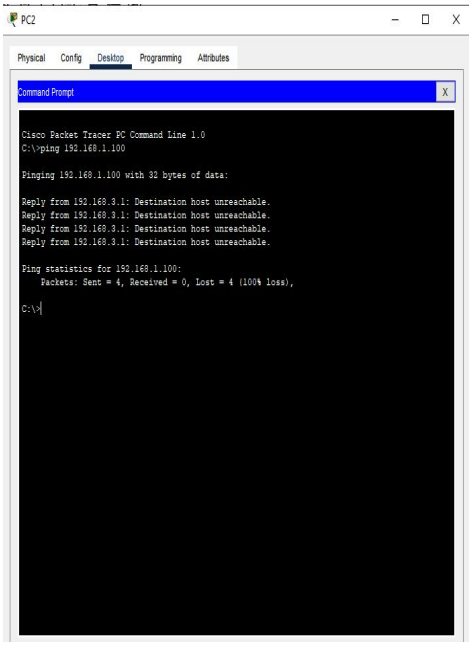


3. Deny Client PC 1, and 2:

- ✓ In same router deny the data from PC 1 and 2 using their source IP
- ✓ PC 1- 192.168.2.100 255.255.255.0
- ✓ PC 2- 192.168.2.101 255.255.255.0

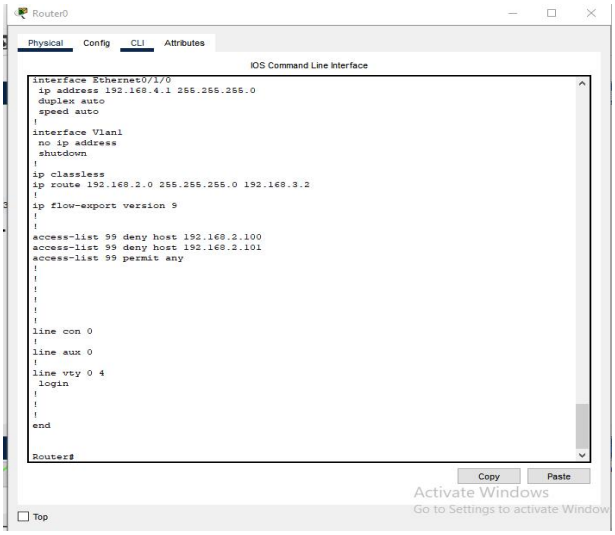


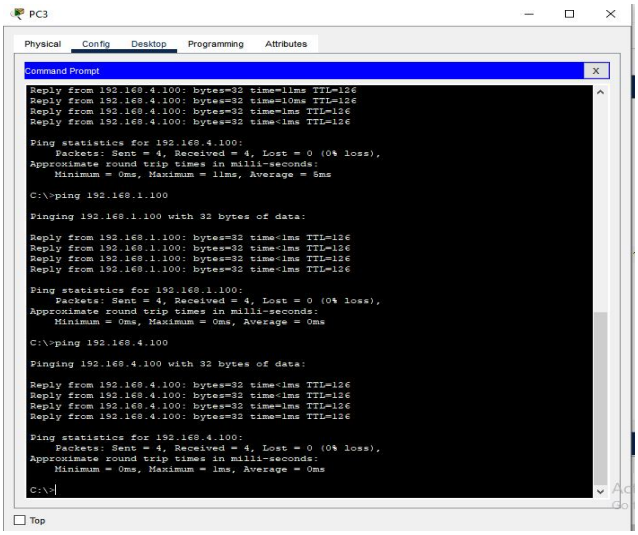




4. Allow PC3 to access all networks:

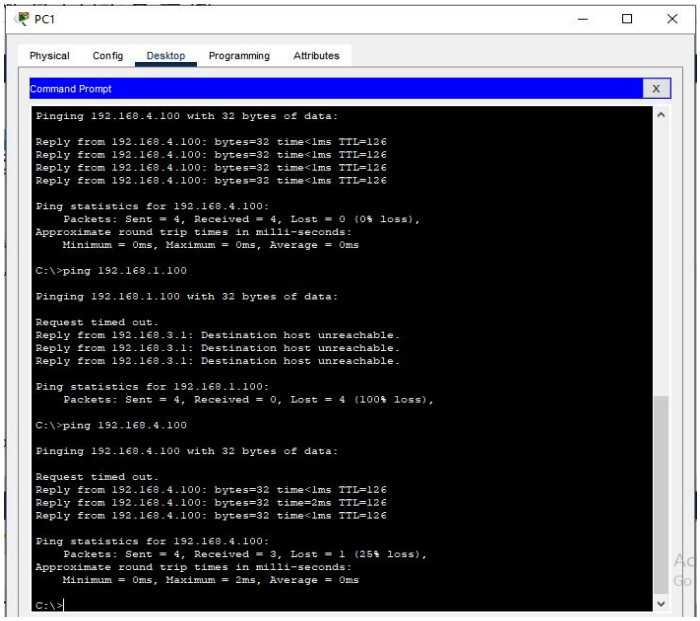
- ✓ In the same run put the permit line
- ✓ "access-list group 1 permit any"
- ✓ PC 3- 192.168.2.102 255.255.255.0

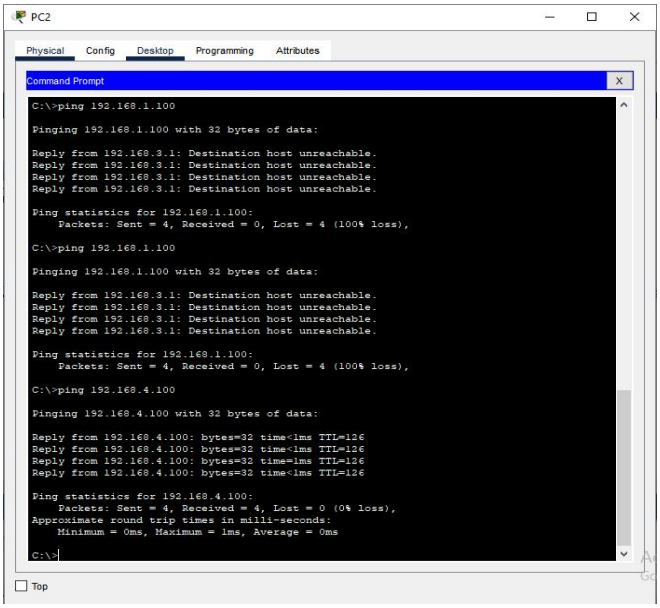


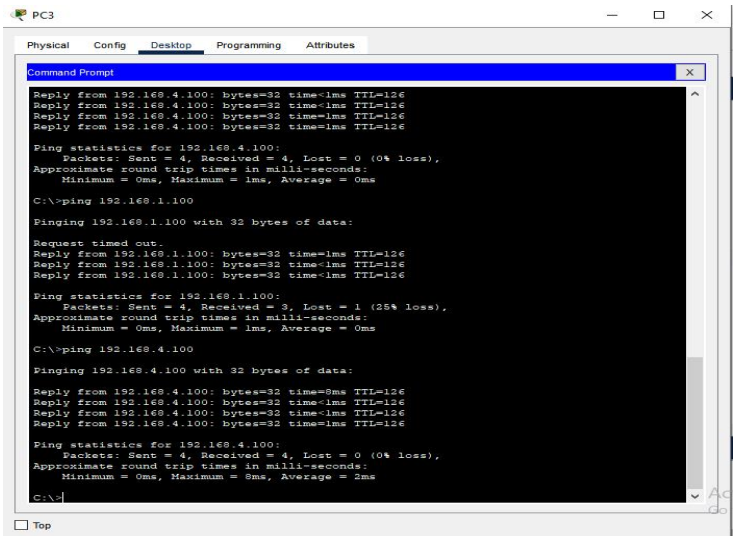


5. Overall Connection:

- ✓ PC1 to PC4
- ✓ PC2 to PC4
- ✓ PC3 to PC0, PC4







Every ping command show proper reply and deny if ip addresses are denied.

E. Conclusion:

The lab exercise on configuring Standard Access Control Lists (ACLs) in Cisco Packet Tracer successfully demonstrated how basic traffic filtering can be implemented using source IP addresses. Through hands-on practice, I learned how to:

- ✓ Create and apply standard ACLs to router interfaces
- ✓ Control network access by permitting or denying specific hosts
- ✓ Verify ACL functionality using simulation tools and packet tracing

This lab reinforced the importance of ACLs in network security and traffic management. While standard ACLs offer limited filtering capabilities compared to extended ACLs, they are effective for simple access control scenarios. Overall, the practical experience enhanced my understanding of ACL syntax, placement, and impact on network behavior.

F. References

- [1] Cisco Systems, Inc., "Configure IP Access Lists," *Cisco Support Documentation*, Oct. 15, 2025. [Online]. Available: *https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/23602-confaccesslists.html*
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